



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Fib series-iterative & recur

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

Sample Output:

Iterative: 0 1 1 2 3 5 8

Recursive: 0 1 1 2 3 5 8

Recursive Calls: 13

Iterative Time: 0.001 ms

Recursive Time: 0.005 ms

Test Case:

Enter N: 1

Output:

0

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main(){
3     int a[]={10,9,7,4,3,1,2};
4     int n=sizeof(a)/sizeof(a[0]);
5     int i,j,max;
6     max=a[0];
7     for(i=1;i<n;i++){
8         if(a[i]>max)
9             max=a[i];
10    }
11    printf("missing elemnet:");
12    for(i=1;i<=max;i++){
13        int found=0;
14        for(j=0;j<n;j++){
15            if(a[j]==i){
16                found=1;
17                break;
18            }
19        }
20        if(found==0)
21            printf("\n %d",i):
```

OUTPUT

```
missing elemnet:
5
6
8
```

INPUT

Your INPUT goes here!



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